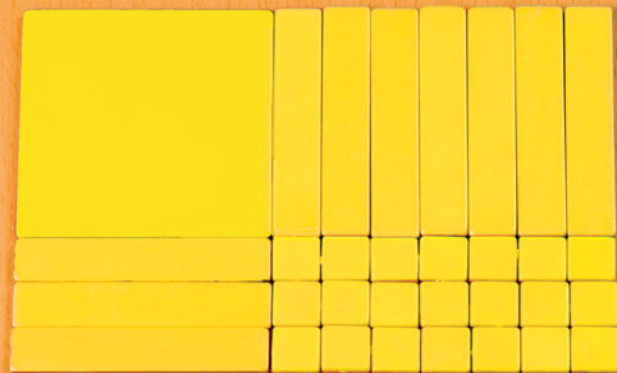


3.5 Polynomials of the Form $x^2 + bx + c$

LESSON FOCUS

Use models and algebraic strategies to multiply binomials and to factor trinomials.



Make Connections

How is each term in the trinomial below represented in the algebra tile model above?

$$(c + 3)(c + 7) = c^2 + 10c + 21$$

Construct Understanding

TRY THIS

Work with a partner.

- Sketch a rectangle to illustrate each product. Write a multiplication sentence that relates the factors to the final product.
 $(c + 4)(c + 2)$ $(c + 4)(c + 3)$ $(c + 4)(c + 4)$ $(c + 4)(c + 5)$
- Describe a pattern that relates the coefficients of the terms in the factors to the coefficients in the product.
- Use the patterns you identified. Take turns to write two binomials, then sketch a rectangle to determine their product.
- Describe a strategy you could use to multiply two binomials, without sketching a rectangle.

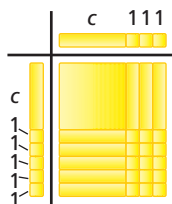
When two binomials contain only positive terms, here are two strategies to determine the product of the binomials.

Use algebra tiles.

To expand: $(c + 5)(c + 3)$

Make a rectangle with dimensions $c + 5$ and $c + 3$.

Place tiles to represent each dimension, then fill in the rectangle with tiles.



The tiles that form the product are: 1 c^2 -tile, 8 c -tiles, and fifteen 1-tiles.

$$\text{So, } (c + 5)(c + 3) = c^2 + 8c + 15$$

Use an area model.

To expand: $(h + 11)(h + 5)$

Sketch a rectangle with dimensions $h + 11$ and $h + 5$.

Divide the rectangle into 4 smaller rectangles and calculate the area of each.

	h	11
h	$(h)(h) = h^2$	$(h)(11) = 11h$
5	$(5)(h) = 5h$	$(5)(11) = 55$

$$\begin{aligned} \text{So, } (h + 11)(h + 5) &= h^2 + 5h + 11h + 55 && \text{Combine like terms.} \\ &= h^2 + 16h + 55 \end{aligned}$$

Note that $(h + 11)(h + 5) = (h + 5)(h + 11)$ since both products represent the area of the same rectangle.

This strategy shows that there are 4 terms in the product. These terms are formed by applying the distributive property and multiplying each term in the first binomial by each term in the second binomial.

$$(h + 11)(h + 5) = h^2 + 5h + 11h + 55$$

When binomials contain negative terms, it is not easy to determine their product using algebra tiles. We cannot determine the product using an area model because a negative area does not exist, but we can use a rectangle diagram. We can also use the distributive property.

Example 1 Multiplying Two Binomials

Expand and simplify.

a) $(x - 4)(x + 2)$ b) $(8 - b)(3 - b)$

SOLUTIONS

a) $(x - 4)(x + 2)$

Method 1

Use a rectangle diagram.

	x	2
x	$(x)(x) = x^2$	$(x)(2) = 2x$
-4	$(-4)(x) = -4x$	$(-4)(2) = -8$

$$\begin{aligned}(x - 4)(x + 2) &= x^2 + (-4x) + 2x + (-8) && \text{Combine like terms.} \\ &= x^2 - 2x - 8\end{aligned}$$

Method 2

Use the distributive property.

$$\begin{aligned}(x - 4)(x + 2) &= x(x + 2) + (-4)(x + 2) \\ &= x(x) + x(2) + (-4)(x) + (-4)(2) \\ &= x^2 + 2x - 4x - 8 && \text{Combine like terms.} \\ &= x^2 - 2x - 8\end{aligned}$$

b) $(8 - b)(3 - b)$

Use the distributive property.

$$\begin{aligned}(8 - b)(3 - b) &= 8(3 - b) + (-b)(3 - b) \\ &= 8(3) + 8(-b) + (-b)(3) + (-b)(-b) \\ &= 24 - 8b - 3b + b^2 \\ &= 24 - 11b + b^2\end{aligned}$$

CHECK YOUR UNDERSTANDING

1. Expand and simplify.

a) $(c + 3)(c - 7)$

b) $(5 - s)(9 - s)$

[Answers: a) $c^2 - 4c - 21$
b) $45 - 14s + s^2$]

How do the constant terms in the binomial factors relate to the middle term and the last term in the trinomial product? How can you use these relationships to determine the product?

Factoring and multiplying are inverse processes. We can use this to factor a trinomial.

When a trinomial contains only positive terms, we may use algebra tiles to factor it.

For example: to factor $v^2 + 12v + 20$, arrange the tiles that represent these terms in a rectangle. Place the v^2 -tile in the top left, then arrange the v -tiles to the right and beneath this tile so the 1-tiles fit in the space that is left.

How does the use of algebra tiles for factoring relate to the area model for multiplication?



$$\text{So, } v^2 + 12v + 20 = (v + 2)(v + 10)$$

We say that the factors of $v^2 + 12v + 20$ are $v + 2$ and $v + 10$.

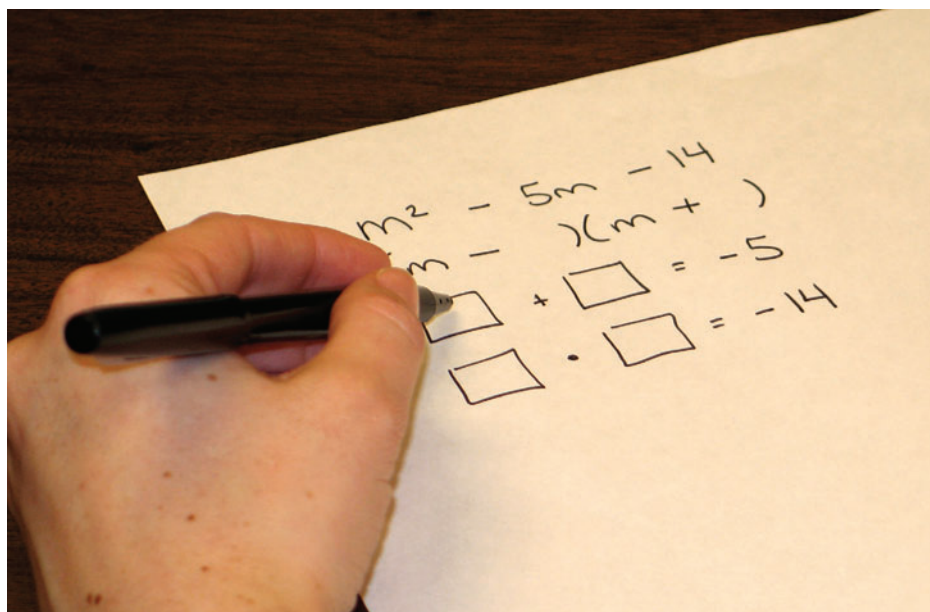
Look at the numbers in the trinomial and the binomial.

$$v^2 + 12v + 20 = (v + 2)(v + 10)$$

12 is the sum of 2 and 10.

20 is the product of 2 and 10.

How could you complete this factorization?



Factoring a Trinomial

To determine the factors of a trinomial of the form $x^2 + bx + c$, first determine two numbers whose sum is b and whose product is c . These numbers are the constant terms in two binomial factors, each of which has x as its first term.

Example 2 Factoring Trinomials

Factor each trinomial.

a) $x^2 - 2x - 8$

b) $z^2 - 12z + 35$

SOLUTION

a) $x^2 - 2x - 8$

Since factoring and expanding are inverse processes, there will be two binomial factors.

They will have the form: $(x + \text{an integer})(x + \text{an integer})$

The two integers in the binomials have a sum of -2 and a product of -8 .

To determine the integers, list pairs of factors of -8 , then add the factors in each pair.

Since the product is negative, the factors have different signs.

Factors of -8	Sum of Factors
$-1, 8$	$-1 + 8 = 7$
$1, -8$	$1 - 8 = -7$
$-2, 4$	$-2 + 4 = 2$
$2, -4$	$2 - 4 = -2$

The factors that have a sum of -2 are 2 and -4 .

Write these integers as the second terms in the binomials.

$$x^2 - 2x - 8 = (x + 2)(x - 4)$$

b) $z^2 - 12z + 35$

Find two integers whose sum is -12 and whose product is 35 .

Since the product is positive, the factors have the same sign.

Factors of 35	Sum of Factors
$1, 35$	$1 + 35 = 36$
$-1, -35$	$-1 - 35 = -36$
$5, 7$	$5 + 7 = 12$
$-5, -7$	$-5 - 7 = -12$

The factors that have a sum of -12 are -5 and -7 .

$$z^2 - 12z + 35 = (z - 5)(z - 7)$$

CHECK YOUR UNDERSTANDING

2. Factor each trinomial.

a) $x^2 - 8x + 7$

b) $a^2 + 7a - 18$

[Answers: a) $(x - 7)(x - 1)$

b) $(a + 9)(a - 2)$]

Does the order in which the binomial factors are written affect the solution? Explain. Why do we list all the factors for the given product rather than all the numbers that have the given sum?

We should always check that the binomial factors are correct by expanding the product.

For *Example 2b*, expand $(z - 5)(z - 7)$.

$$\begin{aligned}(z - 5)(z - 7) &= z(z - 7) - 5(z - 7) \\ &= z^2 - 7z - 5z + 35 \\ &= z^2 - 12z + 35\end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

The trinomials in *Example 2* are written in *descending order*; that is, the terms are written in order from the term with the greatest exponent to the term with the least exponent.

When the order of the terms is reversed, the terms are written in *ascending order*.

Example 3 Factoring a Trinomial Written in Ascending Order

Factor: $-24 - 5d + d^2$

SOLUTIONS

Method 1

$$-24 - 5d + d^2$$

The binomials will have the form:
(an integer + d)(an integer + d)

Find two integers whose product is -24 and whose sum is -5 .

The integers are -8 and 3 .

$$\text{So, } -24 - 5d + d^2 = (-8 + d)(3 + d)$$

Method 2

Rewrite the polynomial in descending order.

$$-24 - 5d + d^2 = d^2 - 5d - 24$$

Find two integers whose product is -24 and whose sum is -5 .

The integers are -8 and 3 .

$$d^2 - 5d - 24 = (d - 8)(d + 3)$$

CHECK YOUR UNDERSTANDING

3. Factor: $-30 + 7m + m^2$

[Answer: $(-3 + m)(10 + m)$]

Is the answer obtained in Method 1 equivalent to that obtained in Method 2? Does the order we write the terms of the binomial matter? Why or why not?

A trinomial that can be written as the product of two binomial factors may also have a common factor.

Example 4

Factoring a Trinomial with a Common Factor and Binomial Factors

Factor.

$$-4t^2 - 16t + 128$$

SOLUTION

$$-4t^2 - 16t + 128$$

The greatest common factor is 4. Since the coefficient of the first term is negative, use -4 as the common factor.

$$-4t^2 - 16t + 128 = -4(t^2 + 4t - 32)$$

Two numbers with a sum of 4 and a product of -32 are -4 and 8 .

$$\text{So, } t^2 + 4t - 32 = (t - 4)(t + 8)$$

$$\text{And, } -4t^2 - 16t + 128 = -4(t - 4)(t + 8)$$

Since factoring and expanding are inverse processes, check that the factors are correct. Multiply the factors.

$$\begin{aligned} -4(t - 4)(t + 8) &= -4(t^2 + 4t - 32) \\ &= -4t^2 - 16t + 128 \end{aligned}$$

This trinomial is the same as the original trinomial, so the factors are correct.

CHECK YOUR UNDERSTANDING

4. Factor.

$$-5h^2 - 20h + 60$$

$$[\text{Answer: } -5(h - 2)(h + 6)]$$

What other ways can you write the trinomial as a product of 3 factors?

When we compare factors of a trinomial, it is important to remember that the order in which we add terms does not matter, so $x + a = a + x$, for any integer a . Similarly, the order in which we multiply terms does not matter, so $(x + a)(x + b) = (x + b)(x + a)$.

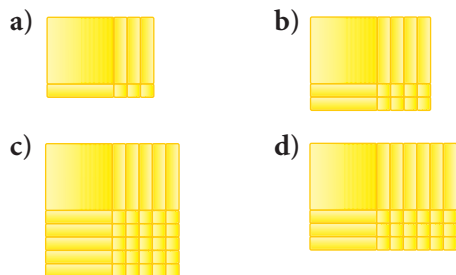
Discuss the Ideas

1. How is representing the product of two binomials similar to representing the product of two 2-digit numbers?
2. How does a rectangle diagram relate binomial multiplication and trinomial factoring?
3. For the multiplication sentence $x^2 + ax + b = (x + c)(x + d)$, what relationships exist among a , b , c , and d ?

Exercises

A

4. Write the multiplication sentence that each set of algebra tiles represents.

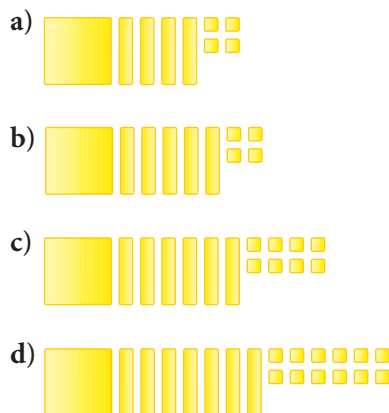


5. Use algebra tiles to determine each product. Sketch the tiles you used.

- a) $(b + 2)(b + 5)$
 b) $(n + 4)(n + 7)$
 c) $(h + 8)(h + 3)$
 d) $(k + 1)(k + 6)$

6. For each set of algebra tiles below:

- i) Write the trinomial that the algebra tiles represent.
 ii) Arrange the tiles to form a rectangle. Sketch the rectangle.
 iii) Use the rectangle to factor the trinomial.



7. a) Find two integers with the given properties.

	a	b	Product ab	Sum $a + b$
i)			2	3
ii)			6	5
iii)			9	10
iv)			10	7
v)			12	7
vi)			15	8

- b) Use the results of part a to factor each trinomial.

- i) $v^2 + 3v + 2$ ii) $w^2 + 5w + 6$
 iii) $s^2 + 10s + 9$ iv) $t^2 + 7t + 10$
 v) $y^2 + 7y + 12$ vi) $h^2 + 8h + 15$

B

8. a) Use algebra tiles to factor each trinomial. Sketch the tiles you used.

- i) $v^2 + 2v + 1$ ii) $v^2 + 4v + 4$
 iii) $v^2 + 6v + 9$ iv) $v^2 + 8v + 16$

- b) What patterns do you see in the algebra tile rectangles? How are these patterns shown in the binomial factors?
 c) Write the next 3 trinomials in the pattern and their binomial factors.

9. Multiply each pair of binomials. Sketch and label a rectangle to illustrate each product.

- a) $(m + 5)(m + 8)$ b) $(y + 9)(y + 3)$
 c) $(w + 2)(w + 16)$ d) $(k + 13)(k + 1)$

10. Copy and complete.

- a) $(w + 3)(w + 2) = w^2 + \square w + 6$
 b) $(x + 5)(x + \square) = x^2 + \square x + 10$
 c) $(y + \square)(y + \square) = y^2 + 12y + 20$

11. Factor. Check by expanding.

- a) $x^2 + 10x + 24$ b) $m^2 + 10m + 16$
 c) $p^2 + 13p + 12$ d) $s^2 + 12s + 20$
 e) $n^2 + 12n + 11$ f) $h^2 + 8h + 12$
 g) $q^2 + 7q + 6$ h) $b^2 + 11b + 18$

- 12.** Expand and simplify. Sketch a rectangle diagram to illustrate each product.
- a) $(g - 3)(g + 7)$ b) $(h + 2)(h - 7)$
 c) $(11 - j)(2 - j)$ d) $(k - 3)(k + 11)$
 e) $(12 + h)(7 - h)$ f) $(m - 9)(m + 9)$
 g) $(n - 14)(n - 4)$ h) $(p + 6)(p - 17)$
- 13.** Find and correct the errors in each expansion.
- a) $(r - 13)(r + 4) = r(r + 4) - 13(r + 4)$
 $= r^2 + 4r - 13r + 52$
 $= r^2 + 9r + 52$
- b) $(s - 15)(s - 5) = s(s - 15) + 15(s + 5)$
 $= s^2 - 15s + 15s + 75$
 $= s^2 + 75$
- 14.** Factor. Check by expanding.
- a) $b^2 + 19b - 20$ b) $t^2 + 15t - 54$
 c) $x^2 + 12x - 28$ d) $n^2 - 5n - 24$
 e) $a^2 - a - 20$ f) $y^2 - 2y - 48$
 g) $m^2 - 15m + 50$ h) $a^2 - 12a + 36$
- 15.** Factor. Check by expanding.
- a) $12 + 13k + k^2$ b) $-16 - 6g + g^2$
 c) $60 + 17y + y^2$ d) $72 - z - z^2$
- 16.** a) Simplify each pair of products.
 i) $(x + 1)(x + 2)$ and $11 \cdot 12$
 ii) $(x + 1)(x + 3)$ and $11 \cdot 13$
 b) What are the similarities between the two answers for each pair of products?
- 17.** Find and correct the errors in each factorization.
- a) $m^2 - 7m - 60 = (m - 5)(m - 12)$
 b) $w^2 - 14w + 45 = (w + 3)(w - 15)$
 c) $b^2 + 9b - 36 = (b + 3)(b - 12)$
- 18.** a) Expand each product, then write it as a trinomial.
 i) $(t + 4)(t + 7)$ ii) $(t - 4)(t - 7)$
 iii) $(t - 4)(t + 7)$ iv) $(t + 4)(t - 7)$
 b) i) Why are the constant terms in the trinomials in parts i and ii above positive?
 ii) Why are the constant terms in the trinomials in parts iii and iv above negative?
 iii) How could you determine the coefficient of the t -term in the trinomial without expanding?

- 19.** Find an integer to replace $\square$ so that each trinomial can be factored.
 How many integers can you find each time?
- a) $x^2 + \square x + 10$
 b) $a^2 + \square a - 9$
 c) $t^2 + \square t + 8$
 d) $y^2 + \square y - 12$
 e) $h^2 + \square h + 18$
 f) $p^2 + \square p - 16$
- 20.** Find an integer to replace $\square$ so that each trinomial can be factored.
 How many integers can you find each time?
- a) $r^2 + r + \square$ b) $h^2 - h + \square$
 c) $b^2 + 2b + \square$ d) $z^2 - 2z + \square$
 e) $q^2 + 3q + \square$ f) $g^2 - 3g + \square$
- 21.** Factor.
- a) $4y^2 - 20y - 56$ b) $-3m^2 - 18m - 24$
 c) $4x^2 + 4x - 48$ d) $10x^2 + 80x + 120$
 e) $-5n^2 + 40n - 35$ f) $7c^2 - 35c + 42$

C

- 22.** In this lesson, you used algebra tiles to multiply two binomials and to factor a trinomial when all the terms were positive.
- a) How could you use algebra tiles to expand $(r - 4)(r + 1)$?
 Sketch the tiles you used. Explain your strategy.
- b) How could you use algebra tiles to factor $t^2 + t - 6$?
 Sketch the tiles you used. Explain your strategy.
- 23.** a) Factor each trinomial.
 i) $h^2 - 10h - 24$
 ii) $h^2 + 10h - 24$
 iii) $h^2 - 10h + 24$
 iv) $h^2 + 10h + 24$
 b) In part a, all the trinomials have the same numerical coefficients and constant terms, but different signs. Find other examples like this, in which all 4 trinomials of the form $h^2 \pm bh \pm c$ can be factored.

Reflect

Suppose a trinomial of the form $x^2 + ax + b$ is the product of two binomials. How can you determine the binomial factors?